

Swiss Institute of Banking and Finance



University of St.Gallen

5. Cost of Capital

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3,130/4,130 Corporate Finance

Course outline

1. General Information
2. Risk and Return
3. Portfolio Theory and CAPM
4. Corporate Valuation
- 5. Cost of Capital**
6. Corporate Financing and the Issuance of Securities
7. Capital Structure
8. Risk Management
9. Conflicts of Interest and Corporate Governance
10. Market Efficiency and Behavioral (Corporate) Finance
11. Empirical Research in Corporate Finance

Agenda

1. Asset betas and the company cost of capital
2. Valuation of the entire firm
 - Using the Weighted Average Cost of Capital (WACC) in DCF valuations
 - Adjusted Present Value
3. Valuation of projects
 - Valuing projects using NPV
 - Bad Outcomes (Appendix A)
 - Certainty Equivalents (Appendix B)
4. An alternative valuation tool: Internal Rate of Return (IRR)
5. The “WACC fallacy”: Empirical evidence

Company cost of capital: WACC

- Definition: The discount rate for the firm's **average-risk** projects.
- The company's cost of capital can be understood as the expected return on a portfolio of **all** the company's existing securities.
 - This portfolio usually includes both debt and equity. It is also referred to as Weighted Average Cost of Capital (WACC).

$$WACC = r_A = \frac{E}{D + E} r_E + \frac{D}{D + E} r_D$$

- E : Market value of the equity (Number of shares · Price per share)
- D : Market value of the debt (often approximated by book value)
- r_E : Cost of equity
- r_D : Cost of debt

Asset betas as a function of equity and debt betas

- Alternatively, we can also use an asset beta to compute the company cost of capital.
- According to the CAPM formula, a firm's cost of capital (r_A) depends on its asset beta and thus on the systematic risk of its existing assets:

$$r_A = r_f + \beta_A (r_M - r_f)$$

- The asset beta can be expressed as the weighted average of the equity and debt beta:

$$\beta_A = \frac{E}{D + E} \beta_E + \frac{D}{D + E} \beta_D$$



IMPORTANT:
D and E are
market values!

Asset betas as a function of equity and debt betas (II)

- Estimating an asset beta requires estimates of the equity and debt betas:

Equity beta

- Use the CAPM:

$$\beta_E = \frac{\text{Cov}[r_E; r_M]}{\text{Var}[r_M]}$$

Debt beta

- For investment grade debt often **assumed** to be zero.
- However, even nearly risk-free bonds have a positive beta.

Determinants of asset betas: General observations

1. Cyclicity

Cyclical firms, i.e., firms whose revenues and earnings strongly depend on the state of the business cycle tend to be high-beta firms. Examples include airlines, luxury products, and construction.

2. Operating leverage (= fixed costs / variable costs of project)

Other things being equal, a project that has higher relative fixed costs and/or higher commitments to invest will have a higher project beta (see BMAE, p. 259 f. for a formal derivation).

3. Time horizon of the project

Projects with long-term cash flows are more sensitive to changes in the risk-free rate or market risk premium.

Cost of equity: Who uses the CAPM?

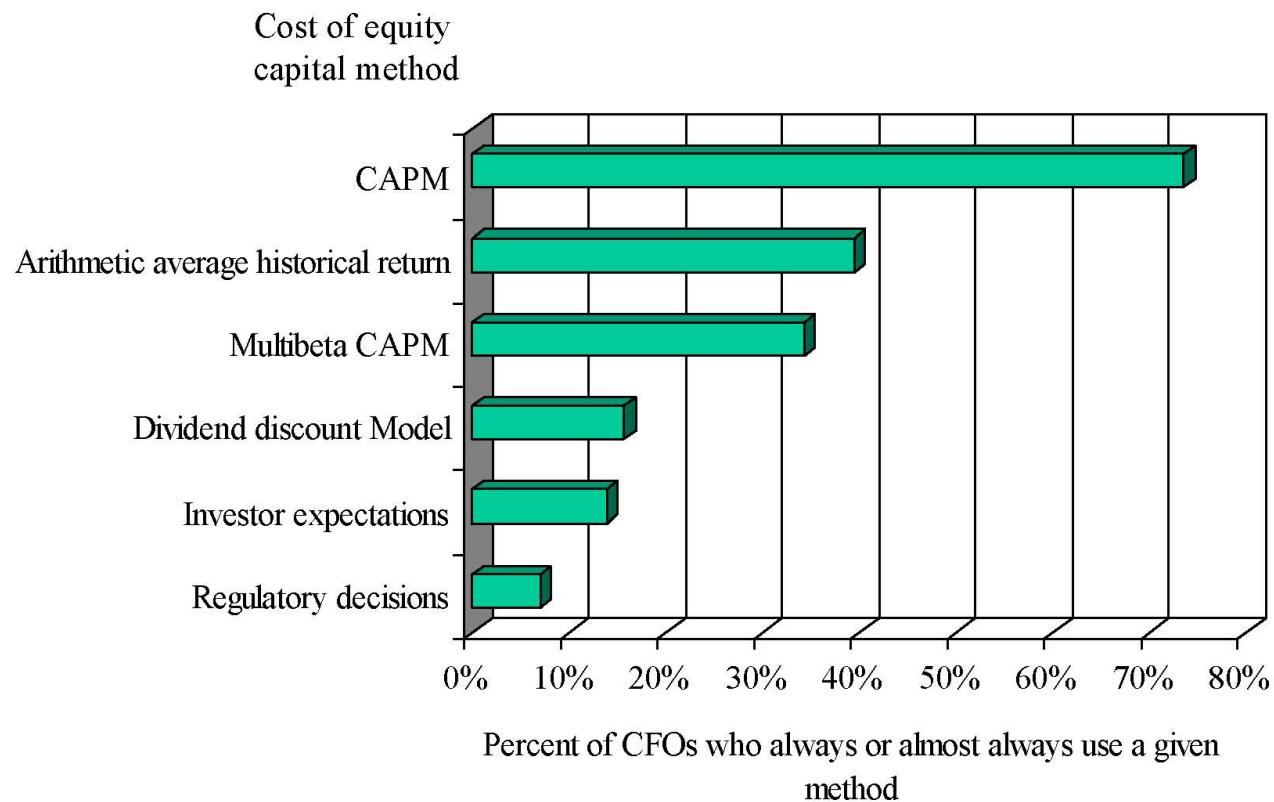


Fig. 3. Survey evidence on the popularity of different methods to calculate the cost of equity capital. We report the percentage of CFOs who always or almost always use a particular technique. CAPM represents the capital asset pricing model. The survey is based on the responses of 392 CFOs.

Source: Graham and Harvey (2001)

Cost of debt

- Debt differs from equity in one remarkable way:
 - Equity: shareholders are entitled to receive what is left after all claimants are paid off (“residual claimants”)
 - Upside potential
 - Debt: bondholders are entitled to receive a promised payment each year (interest) and repayment of the principal at maturity
 - No upside potential, only downside risk
- A major source of risk of debt is default (repayment) risk
- Borrowers with higher default risk should pay higher interest rates on debt

Cost of debt (II)

- Default risk is the probability that companies or individuals will be unable to make the required payments on their debt obligations.
- Bond rating agencies base their creditor ratings largely on an analysis of the level and trend of the issuer's financial ratios, e.g.:
 - coverage ratios (company earnings to fixed costs),
 - leverage ratio (debt-to-equity ratio),
 - liquidity ratios (e.g., current, quick ratio),
 - profitability ratios (e.g., return on equity, return on assets), and
 - cash flow-to-debt ratio.
- An analysis of such ratios allows an assessment of credit risk also for non-rated bonds / debt.

Cost of debt (III)

- **Bond default risk**, so-called credit risk, is measured by the three major bond rating agencies Moody's, Standard & Poor's (S&P), and Fitch. The ratings issued by these companies **estimate** a bond's riskiness (or safety).

Moody's	S&P and Fitch
Investment grade bonds	
Aaa	AAA
Aa	AA
A	A
Baa	BBB
Junk bonds	
Ba	BB
B	B
Caa	CCC
Ca	CC
C	C

Source: BMAE (2022, p. 700, Table 24.1)

Cost of debt (IV)

Higher ratings are associated with **lower credit spreads** and thus lower costs of debt.

Reuters Corporate Spreads for Industrials (12.06.2019)							
Rating	1 yr	2 yr	3 yr	5 yr	7 yr	10 yr	30 yr
AAA	11	22	28	37	45	60	88
AA	25	36	42	53	63	78	105
A	38	52	58	70	83	101	121
BBB	63	82	92	112	134	160	182
BB	143	195	224	268	305	339	384
B	215	274	308	357	400	432	478

Spread values represent basis points (bps)* over a US Treasury security of the same maturity, or the closest matching maturity.

*100 bps = 1%

Source:
Bloomberg

Cost of debt (IV)

Higher ratings are associated with **lower credit spreads** and thus lower costs of debt.

Reuters Corporate Spreads for Industrials (31.12.2022)							
Rating	1 yr	2 yr	3 yr	5 yr	7 yr	10 yr	30 yr
AAA	27	32	37	48	55	67	83
AA	19	31	47	81	101	119	121
A	33	49	60	86	95	118	124
BBB	145	119	109	138	157	181	153
BB	321	315	315	338	379	421	608
B	502	510	520	541	598	660	799

Spread values represent basis points (bps)* over a US Treasury security of the same maturity, or the closest matching maturity.

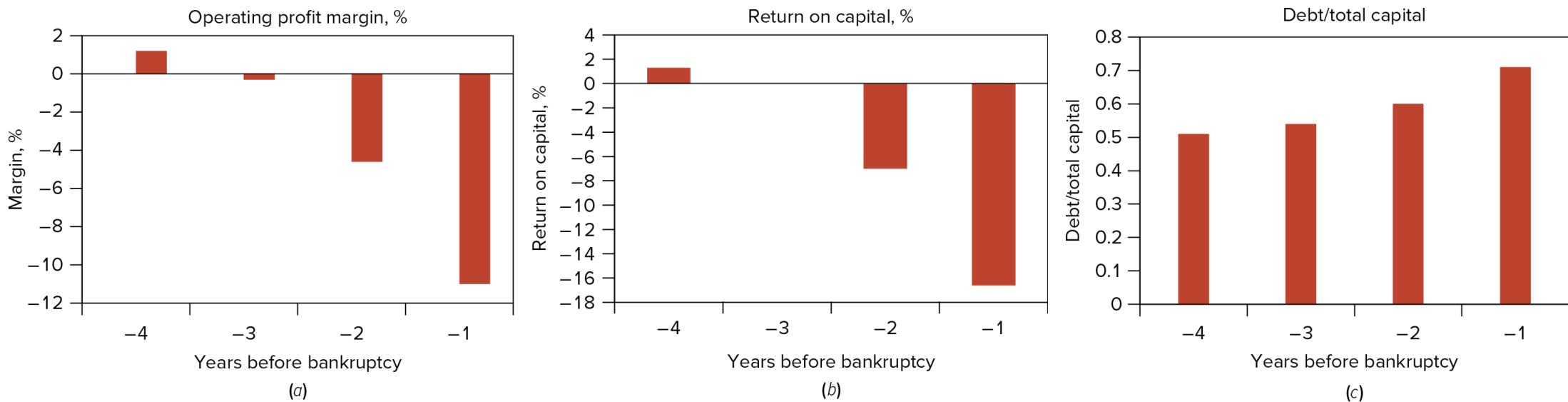
*100 bps = 1%

Source:
Bloomberg

Default risk

Characteristics of failing firms

Do such financial ratios work as bankruptcy predictors?



Source: BMAE (2022, p. 703, Figure 24.4)

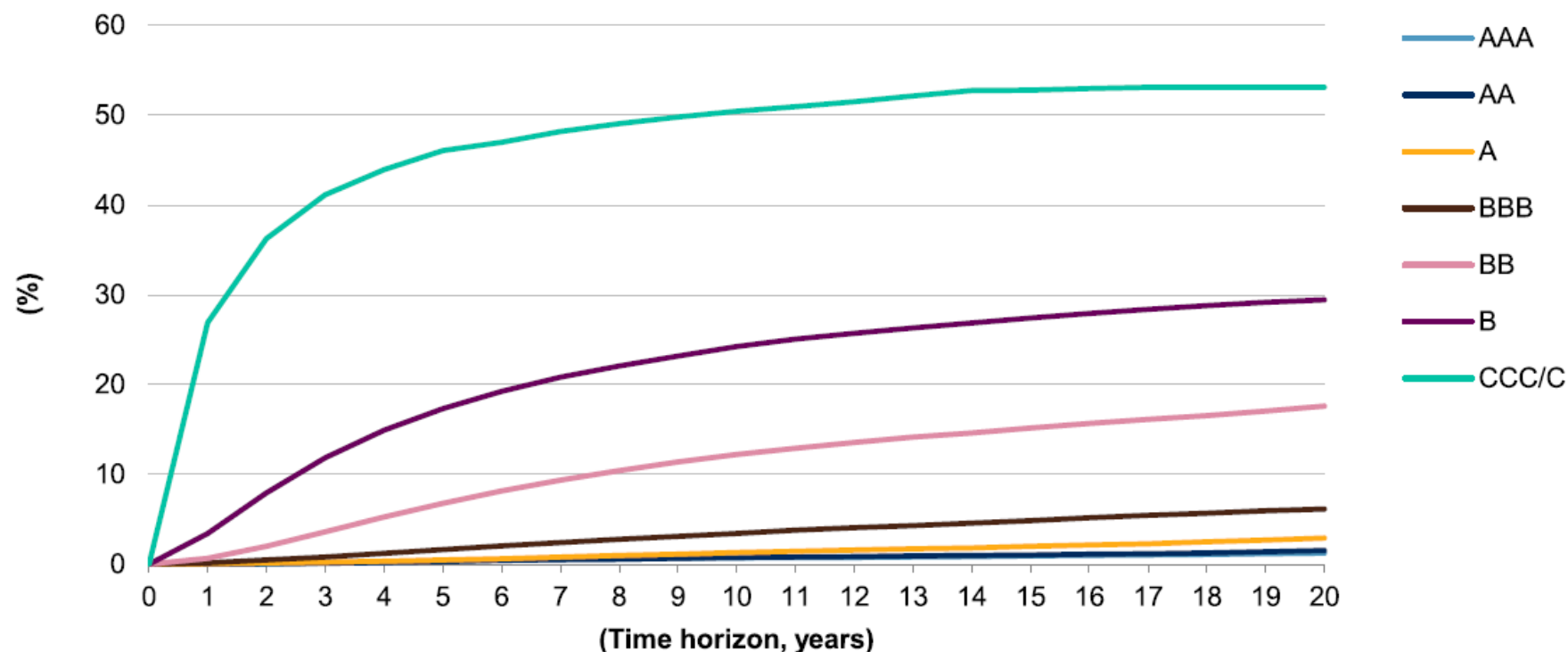
Beaver et al. (2005), "Have Financial Statements Become Less Informative? Evidence from the Ability of Financial Ratios to Predict Bankruptcy", *Review of Accounting Studies* (10), 93-122

Default risk

Do bond ratings predict the probability to default?

- Bond ratings reflect the probability to default.

Global Corporate Average Cumulative Default Rates By Rating (1981-2018)



Sources: S&P Global Fixed Income Research and S&P Global Market Intelligence's CreditPro®.

S&P Global Ratings: 2018 Annual Global Corporate Default And Rating Transition Study

Cost of debt (V)

- In practice, the cost of debt r_D is estimated as

$$r_D = r_f + \text{default spread}$$

- Simple assumption: For investment grade debt, *default spread* = 0

$$r_D = r_f$$

- Better: Estimate default spreads

1. Build a sample of bonds comparable to the debt we want to value:

- Similar industry, similar ratings or financial ratios, similar maturity etc.
- Focus on liquid bonds without features such as convertibility etc.

2. Estimate average yield of sample:

- Estimate yield to maturity of the bonds
- Apply weighting if similarity among bonds is not given

Weighted Average Cost of Capital (WACC)

- The company cost of capital, or WACC, as shown on slide 4 ignores taxes.
- However, **interest expenses are normally tax-deductible**. For every dollar in interest the firm pays, it receives a tax deduction of T_c dollars (T_c is the marginal corporate tax rate). Therefore, the cost of debt has to be adjusted:

$$WACC = \frac{E}{D + E} r_E + \frac{D}{D + E} r_D \cdot (1 - T)$$

Weighted Average Cost of Capital (WACC) (II)

- WACC covers the overall risk of the firm and the tax shield of debt.
- But:
 - The WACC formula is correct only for projects that are just like the firm undertaking them (same risk and debt ratio). Hence, the WACC formula works for the “average” project.
 - WACC is incorrect for projects which are riskier or safer than the average of the firm’s existing assets. It is also incorrect for projects which would change the firm’s (target) debt ratio.
 - WACC is based on a firm’s current characteristics, but managers use it to discount future cash flows. This is only correct if the firm’s business risk and debt ratio are expected to remain constant.

WACC and DCF

Discounted Cash Flow (DCF) Method:

$$PV_0 = \frac{FCF_1}{1+r} + \frac{FCF_1}{(1+r)^2} + \dots + \frac{FCF_H}{(1+r)^H} + \frac{PV_H}{(1+r)^H} = \sum_{t=1}^H \frac{FCF_t}{(1+r)^t} + \frac{PV_H}{(1+r)^H}$$

See Lecture 4, slide 28

- Substitute WACC for r to estimate the firm value.
- Use the FCFs of the entire firm:

**Free cash flow = Profit after tax
+ depreciation
– investment in fixed assets
– investment in working capital**

- Subtract the value of debt from the firm value to get the firm's equity value.

WACC and DCF (II)

WACC – Warning

- Check validity of results by looking at “comparable” firms in terms of:
 - **Multiples** (such as price-earnings, or market-to-book),
 - **Revenue** figures and **expected growth**.

⇒ Validate r_E using stock market data of these “comparables”.
- Most of the firm value stems from the terminal value. Growth rate and profitability assumptions must be made carefully and prudently.

WACC and DCF (III)

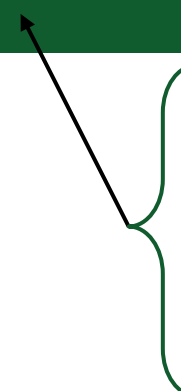
Adjusting WACC when leverage or business risk differs

- Leverage might differ between the company and its new project.
- For example, assume the project has a higher debt ratio:
 - Financing weights will change,
 - Both cost of equity and cost of debt will increase since financial risk will be higher,
 - The WACC will, however, decrease overall due to the tax shield on debt interest payments.
- Note: For WACC to be valid, the debt ratio must be held constant and **actively rebalanced** if equity value increases or falls!

Valuation with taxes: Adjusted Present Value

- We just learned that one key assumption of using the WACC approach in valuation is that the firm actively **rebalances** in order to maintain its debt ratio. If this is not the case, the APV approach will be more appropriate (than DCF).
- The APV method calculates a series of present values: First, the PV if the firm was all-equity financed. Second, the PVs of the costs and benefits associated with the firm's financing.

$$\text{APV} = \text{base-case NPV} + \text{sum of PVs of financing side-effects}$$



- + Tax shield
- + Loan subsidies
- Issuance costs
- Costs of fin. distress

Valuation w. taxes: Adjusted Present Value – Example

Glory Inc. (see company characteristics below) plans to undertake a new investment project (also detailed below).

What is the value of the new project?

Glory Inc. Case		
Balance sheet	Costs / Taxes	Investment project
MV assets = 3,000 Debt = 1,000 MV equity = 2,000	$r_D = 5\%$ $r_E = 11\%$ $T = 35\%$	Investment = 150 $CF_{\text{pre-tax}} = 19.43$ (accrues for infinity) same risk structure

Valuation w. taxes: Adjusted Present Value – Example (II)

Calculation of the cost of capital (after tax):

$$WACC_{after\ tax} = \frac{1000}{3000} 0.05(1 - 0.35) + \frac{2000}{3000} 0.11 \approx 0.0842 = 8.42\%$$

In the case of a constant debt ratio, **the risk of the tax shield is the same as the risk of the project.**

The tax shields have to be discounted at the opportunity cost of capital, r , which is the **unlevered WACC (ignoring taxes)**:

$$r = \frac{1000}{3000} 0.05 + \frac{2000}{3000} 0.11 = 0.09 = 9\%$$

Valuation w. taxes: Adjusted Present Value – Example (III)

First step

Calculating the base-case NPV

$$\text{Base-case NPV} = -I + \frac{CF_{pre-tax} \cdot (1-T)}{WACC_{pre-tax}} = -\$150 + \frac{\$19.43 \cdot (1-0.35)}{0.09} = -\$9.7$$

We discount the after-tax project cash flows of \$12.63 at the opportunity cost of capital of 9% and subtract the \$150.00 outlay.

⇒ Hence, ignoring the tax shield, we would **not** undertake the project!

Valuation w. taxes: Adjusted Present Value – Example (IV)

Second step

Calculating the tax shield

The project supports \$50 of debt.

$$\text{Annual tax shield} = T \cdot r_D \cdot D = 0.35 \cdot 0.05 \cdot \$50 = \$0.875$$

If the firm is **constantly rebalancing its debt ratio**, we **discount at the opportunity cost of capital, r** , of 9%.

$$PV(\text{tax shields, debt rebalanced}) = \frac{\text{annual tax shield}}{WACC_{pre-tax}} = \frac{\$0.875}{0.09} = \$9.7$$

APV is the sum of the base-case value and PV tax shield.

$$APV = \text{base-case NPV} + PV \text{ tax shield} = -\$9.7 + \$9.7 = \$0$$

⇒ Holding the debt ratio **constant**, APV and WACC yield the **same** results.

Valuation w. taxes: Adjusted Present Value – Example (V)

Alternative:

Holding the amount of debt fixed at \$50.

In this case, we can assume that the **risk of the tax shield is the same as the risk of debt** and we discount at the 5% debt rate.

$$PV \text{ (tax shields, debt fixed)} = \frac{\text{annual tax shield}}{r_D} = \frac{\$0.875}{0.05} = \$17.5$$

$$APV = \text{base-case NPV} + PV \text{ tax shield} = -\$9.7 + \$17.5 = \$7.8$$

⇒ Now the project is more attractive. With **fixed debt**, the interest tax shield is safer and therefore worth more.

Valuation w. taxes: Adjusted Present Value – Example (VI)

Variation

Finance by issuing \$100 of equity and \$50 of debt

Issue cost of 6% for equity = \$6

Issue cost of 2% for debt = \$1

Assume that debt is fixed and therefore the interest tax shields are worth \$17.5.

$$APV = APV_{old} - \text{issue costs} = \$7.8 - \$6 - \$1 = \$0.8$$

APV – Critical reflection

- Most useful when side-costs are numerous and important
- Projects with commonly high leverage
- Decreasing debt ratio over time
- Subsidized financing packages at low interest rates, etc.

Valuation of projects – Project cost of capital

- To value a project we need to use the expected return for projects with **similar risk characteristics**. The company cost of capital works only for projects that are carbon copies of the firm.
- Imagine that a coal-mining firm wants to assess the risk of investing into real estate. The beta of coal-mining will be of little help here. You would rather have to look at REITs (listed real estate firms) to know the beta of real estate.
 - REITs will be used as a “comparable”.
- The best comparison for one particular line of business will be to consider “**pure plays**” in that line of business.

Valuation of projects – Pure plays

- Find firms that are “pure plays”:
 - in the same line of business as the project considered,
 - not involved in any other type of business, and
 - publicly traded.
- The cost of capital (or asset beta) of a pure play best reflects the industry-specific costs of capital, i.e., the cost of capital (asset beta) of the project you are considering.
- Pure plays are sometimes also called focused firms. Firms that are not pure plays are also called conglomerates.

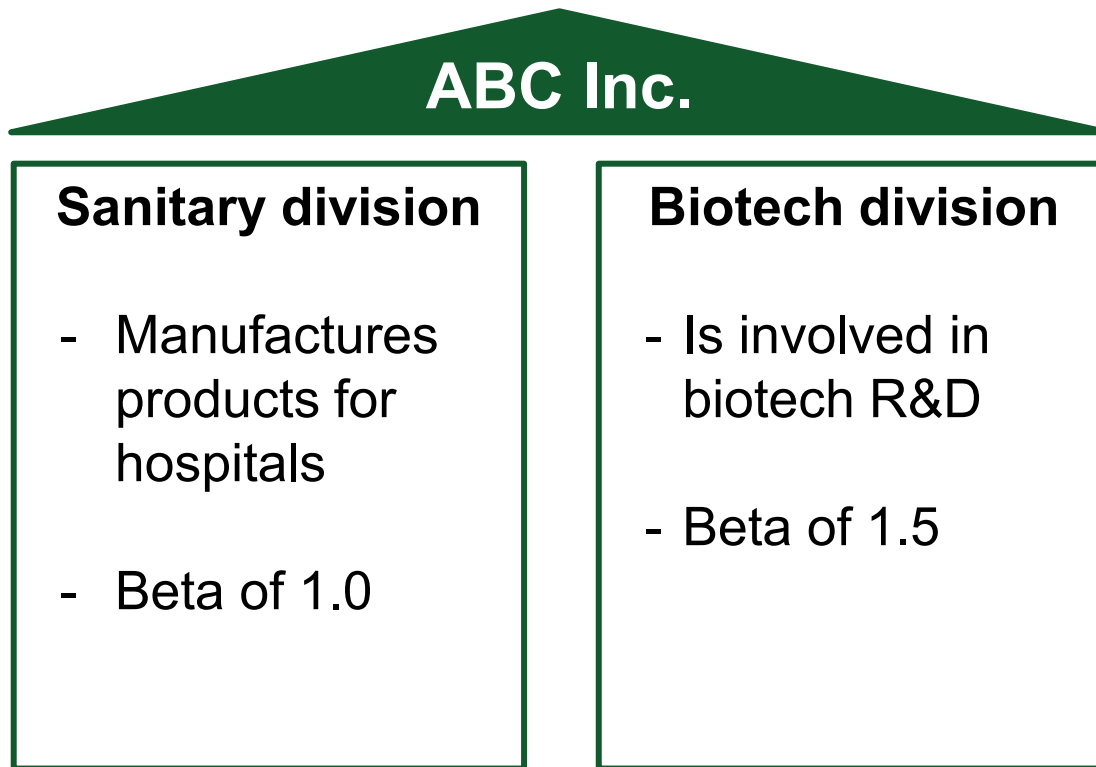
Valuation of projects – Pure plays (II)

- How do we find companies that are focused in the same "line of business" as the project we want to evaluate?
- Usually, we can rely on “segment reporting” to differentiate between focused companies and conglomerates.
- Segments (lines of business) are usually differentiated by some form of industry classification (SIC, NAICS, FF12, FF48).
- Problems of using such industry codes:
 - The assignment of standardized industry codes often seems arbitrary.
 - Segment reporting by companies is subject to significant leeway.
 - Depending on the project, common sense and industry knowledge are superior to any standardized industry classification.

Valuation of projects – Pure plays (Example)

- What would be “pure plays” for VAT Group AG (see last class)?
- The SIC code of VAT Group is 3491. These digits stand for:
 - 3... = Manufacturing
 - 34.. = Fabricated Metal Products, except Machinery and Transportation Equipment
 - 349. = Miscellaneous Fabricated Metal Products
 - 3491 = Industrial Valves
- Examples of pure plays are: Asco Valve Inc. (US), Mercer Valve Co., Inc. (US), Atam Valves Ltd (India), HS Valve Co Ltd (South Korea), Goodwin PLC (UK)
- Note: To address the shortcomings of standardized industry codes, academic research increasingly uses text-based industry classification systems, for example using the similarity of product descriptions in annual reports (e.g., Hoberg and Phillips, 2016).

Valuation of projects – Example



A	P
Sanitary division	Equity
50%	
.....	
Biotech division	
50%	100%

Assume: Market value = Book value

- The risk-free rate is 5%, and the market risk premium ($r_m - r_f$) is 8.4%.

Valuation of projects – Example (II)

- Assume that the company can choose between two mutually exclusive projects:

Project A in the sanitary products division				
Year	0	1	2	3
Cash flows	-100	50	50	26.174

Project B in the biotech division				
Year	0	1	2	3
Cash flows	-50	25	20	22.458

- Which project should the firm choose?

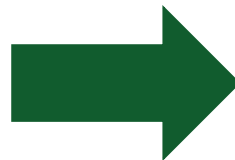
Valuation of projects using net present values

Present value (PV)

The **present value** PV_0 equals the sum of all discounted payoffs:

$$PV_0 = \sum_{t=1}^T \frac{C_T}{(1+r)^T}$$

Cost of the
initial
investment



Net present value (NPV)

Investment decisions are generally based on the **net present value (NPV)**, which subtracts the required investment from the present value.

$NPV = PV - \text{required investment}$

$$NPV = C_0 + \sum_{t=1}^T \frac{C_T}{(1+r_T)^T}$$

Valuation of projects using net present values (Example)

- Since we have to discount each project with its appropriate discount rate, using the company beta may lead to wrong decisions.

Sanitary cost of capital

$$r_{San} = 5 + 1.0 \cdot 8.4 = 13.4\%$$

Biotech cost of capital

$$r_{Bio} = 5 + 1.5 \cdot 8.4 = 17.6\%$$

Company Beta

$$\beta_{ABC} = \frac{1 + 1.5}{2} = 1.25$$

Company cost of capital

$$r_{ABC} = 5 + 1.25 \cdot 8.4 = 15.5\%$$

Project A in the sanitary products division

Year	0	1	2	3
Cash flow	-100	50	50	26.174
NPV (r_{ABC})	-2.24	Wrong!		
NPV (r_{San})	0.92	Correct!		

Project B in the biotech division

Year	0	1	2	3
Cash flow	-50	25	20	22.458
NPV (r_{ABC})	1.21	Wrong!		
NPV (r_{Bio})	-0.47	Correct!		

Valuation of projects with Internal Rate of Return (IRR)

The **internal rate of return (IRR)**, or **yield (y)**, is the discount rate at which the net present value is equal to zero. A simple investment rule is then: Accept investments that offer an IRR in excess of their opportunity cost of capital (i.e., the discount rate).

$$\text{NPV} = C_0 + \frac{C_1}{1+y} + \frac{C_2}{(1+y)^2} + \dots + \frac{C_N}{(1+y)^N} = 0$$

The two-period case

$$0 = C_0 + \frac{C_1}{1+y} + \frac{C_2}{(1+y)^2}$$

$$0 = C_0 + x \cdot C_1 + x^2 \cdot C_2 = 0$$

$$\text{with } x = \frac{1}{1+y}$$



$$x = \frac{-C_1 + \sqrt{C_1^2 + 4 \cdot C_0 \cdot C_2}}{2 \cdot C_2}$$

$$\Rightarrow y = \frac{1}{x} - 1$$

In general, there are three ways to calculate the IRR: (1) graphical method, (2) trial and error method, and (3) using a financial calculator.

Valuation of projects with IRR (Example)

If the internal rate of return is larger than the required rate of return (i.e., cost of capital), we choose the project.

Sanitary cost of capital

$$r_{San} = 5 + 1.0 \cdot 8.4 = 13.4\%$$

Biotech cost of capital

$$r_{Bio} = 5 + 1.5 \cdot 8.4 = 17.6\%$$

Company Beta

$$\beta_{ABC} = \frac{1 + 1.5}{2} = 1.25$$

Company cost of capital

$$r_{ABC} = 5 + 1.25 \cdot 8.4 = 15.5\%$$

Project A in the sanitary products division

Year	0	1	2	3
Cash flow	-100	50	50	26.174
IRR	14%	IRR < r_{ABC} wrong!		

$$IRR > r_{San}$$

Project B in the biotech division

Year	0	1	2	3
Cash flow	-50	25	20	22.458
IRR	17%	IRR > r_{ABC} wrong!		

$$IRR < r_{Bio}$$

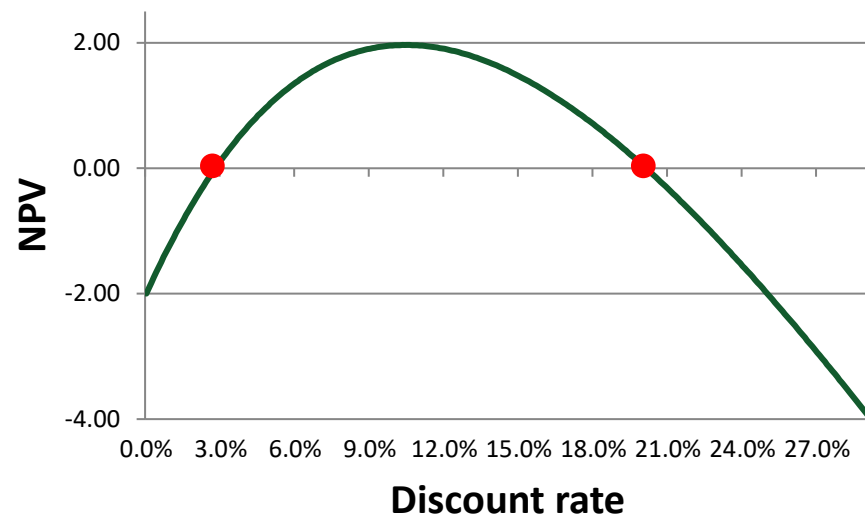
Valuation of projects with IRR

Pitfalls (I)

- Suppose there is a third project:

Project C					
Year	0	1	2	3	4
Cash flow	-100	80	80	80	-142

- What is the correct internal rate of return (IRR)?



IRR1 = 2.8%

IRR2 = 20.1%

For cash flows with more than one change in sign, there can be **multiple IRRs**.

Valuation of projects with IRR

Pitfalls (II)

- The IRR method sometimes **ignores the magnitude of a project**. This applies, in particular, to mutually exclusive projects meaning the firm can only accept one out of two (or more) projects.

Cash Flows, \$				
Project	C_0	C_1	IRR (%)	NPV at 10%
D	−10,000	+20,000	100	+ 8,182
E	−20,000	+35,000	75	+11,818

Source: BMAE (2022, p. 131)

- For **mutually exclusive projects**, accept the project with the higher NPV (which needs not be the one with a higher IRR).

The “WACC fallacy”

A large-sample empirical study

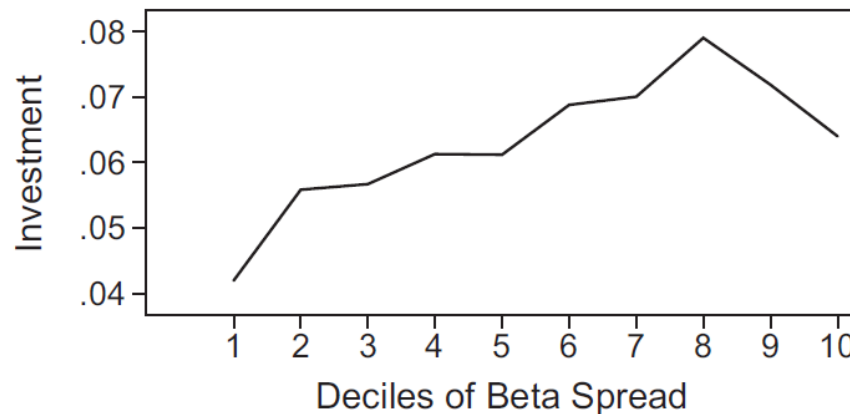
Are the WACC assumptions a problem in the real world?

- Survey evidence suggests that many firms use only one discount rate to value all of their projects.
- Krüger, Landier, and Thesmar (2015) are the first to use large-sample empirical data to test the WACC fallacy and its economic consequences.
- Of course, we cannot observe the discount rate used by thousands of listed US firms.
- Test I: Do diversified firms invest less in low beta and more in high beta divisions (controlling for standard determinants of investment, such as growth options)?

The “WACC fallacy” (II)

A large-sample empirical study

- Results:



Average non-core division investment (vertical axis) by decile of the spread between the industry-level asset beta of the non-core division and that of the core division (horizontal axis).

- Test II: Is there cross-sectional evidence consistent with such a mechanism?
- Results: Firms for which the WACC mistake is more costly...:
 - The non-core division is large
 - The CEO has sizable ownership
 - The within-firm diversity in cost of capital is high...are less likely to do the WACC mistake.

The “WACC fallacy” (III)

A large-sample empirical study

- Test III: What is the present value loss of doing the WACC mistake? They look at diversifying acquisitions as this type of investment can be easily observed, is usually large, and is directly assessed by the market.
- Specific test: Do acquirers with lower cost of capital than that of the target overpay?
- Results:
 - In an event study, the announcement return associated with this type of acquirer is significantly lower by about -0.8% versus other acquirers.
 - This corresponds to about 8% of the deal value (or \$16m per deal).

Key takeaways

1. The (after-tax) **WACC** depends on the average risk of the company's assets, financing structure, and taxes.
2. We can use **WACC** in a **DCF analysis** to find the value of a firm.
3. The **cost of capital for a project** should be estimated on the project, not the firm level, using “pure plays”.
4. An alternative valuation tool is **IRR**. However, there are several **pitfalls** in using IRR. If applied correctly, it is equivalent to using NPV (based on DCF and WACC).
5. **Empirical evidence** suggests that firms make **investment mistakes** by using the wrong cost of capital.

Mandatory readings

- Brealey, R., Myers, S., Allen, F., and A. Edmans, 2022, Principles of Corporate Finance, 14th Edition, New York: McGraw-Hill, Chapter 4 (Valuing Stocks), Chapter 5 (Net Present Value and Other Investment Criteria), Chapter 6 (Making Investment Decisions with the Net Present Value Rule), & Chapter 9 (Risk and the Cost of Capital).

Additional readings

- Graham, J.R., and C.R. Harvey, 2001, The theory and practice of corporate finance: Evidence from the field, Journal of Financial Economics 61, 187-243.
- Hoberg, G., and G. Phillips, 2016, Text-based network industries and endogenous product differentiation, Journal of Political Economy 124, 1423-1465.
- Krüger, P., Landier, A., and D. Thesmar, 2015, The WACC fallacy: The real effects of using a unique discount rate, Journal of Finance 70, 1253-1285.

Appendix A: Allowing for bad outcomes

- Make **unbiased** forecasts of the project's cash flows:
 - An unbiased forecast is the sum of the probability-weighted cash flows.
 - Remember to use **expected CFs** and account for **all** risks (including diversifiable risk and market risk)!
- Do not consider risk that is diversifiable, such as hazards, in the beta calculation or discount rates.
- Estimate whether a **diversified** investor would see the project as more or less risky than the average project.
- Avoid “**fudge factors**” in discount rates
 - Do not adjust discount rates using a rule of thumb, instead adjust CFs.



Allowing for bad outcomes (II)

Example (I)

- Project Z will produce just one cash flow, forecasted at \$1 million at year 1. It is regarded as average risk, suitable for discounting at a 10% company cost of capital.

Possible Cash Flow	Probability	Probability-Weighted Cash Flow
1,200,000	0.25	300,000
1,000,000	0.50	500,000
800,000	0.25	200,000
Unbiased Forecast		1,000,000



$$PV = \frac{1,000,000}{1.1} = \underline{\underline{\$909,100}}$$

(BMAE, 2022, example 9.3, p. 261 ff.)

- But now you discover that the company's engineers are behind schedule in developing the technology required for the project.

Allowing for bad outcomes (III)

Example (II)

- They are confident it will work, but they admit to a small chance (10%) that it will not (cash flow equals 0). You still see the most likely outcome as \$1 million, but you also see some chance that project Z will generate zero cash flow next year.

Possible Cash Flow	Probability	Probability-Weighted Cash Flow
1,200,000	0.225	270,000
1,000,000	0.45	450,000
800,000	0.225	180,000
0	0.1	0
Unbiased Forecast		900,000

(BMAE, 2022, example 9.3, p. 261 ff.)



$$PV = \frac{900,000}{1.1} = \underline{\underline{\$818,000}}$$

Appendix B: Certainty equivalents

- Assume ABC Railroad Works Inc. is building a rail route that it plans to sell after one year for USD 1,000,000. The cash flow is uncertain carrying the same risk as the market, i.e., $\beta = 1$.
- If $r_f = 3\%$ and the market risk premium is at 7% you would be discounting at a rate of 10%.
- Calculating the Present Value at the risk adjusted rate as usual:

$$1,000,000/1.10 \approx \underline{909,091}$$

Certainty equivalents (II)

- Suppose DEF Inc. approaches you and offers to buy the rail route in one year for a price fixed today.
- Which (lower) price would you agree upon when any uncertainty is removed? This is the **certainty equivalent**:

$$PV = \frac{\text{certain cash flow}}{1 + r_f}$$

$$\text{Certain cash flow} \approx \underline{936.364}$$

- Using the risk-adjusted rate compensates for both: time and risk.
- The certainty equivalent method makes separate adjustments for time and risk.